ACHILLES: A Machine Learning Framework for Explainable and Generalized Automotive Intrusion Detection System

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***Abstract*—This paper addresses the need for an explainable and generalized intrusion detection system (IDS) for the in-vehicle networks (IVNs). While machine learning (ML)-based IDS solutions show promising performance, there are still some challenges, such as the lack of trustworthiness and scarcity of attack representing data, hindering their adoption in the automotive cybersecurity. To address these issues, this paper proposes a centralized ML model training and decentralized execution-based framework, namely ACHILLES, that facilitates an explainable and generalizable automotive IDS. Under ACHILLES, different ML models can be trained centrally to enhance decentralized and onboard intrusion detection performance with multiple automotive datasets. In addition, we generate standard feature formats to assess the ML model's generalization efficacy, where the quality of generalization and explainability is evaluated with SHapley Additive exPlanations (SHAP) by identifying the importance of the feature. We also propose a meta-learning scheme to construct suitable ML models trained by the proposed standard feature formats. The proposed feature format exhibits significant performance gain during ML model training and testing with four state-of-the-art controller area network (CAN)-bus datasets containing real, advanced attacks. The experimental results indicate that developing ML models using the generated generalized features and the meta learning-based model building process leads to enhanced performance. In particular, under the dataset cross train-test setting, the proposed feature format enhances the average accuracy by 40.1% for the baseline model, 32.4% for the meta-learned DNN, and 23.6% for the meta-learned Random Forest, compared with the baseline feature format.**

***Index Terms*—In-vehicle networks (IVNs), automotive intrusion detection, generalizability, explainable AI (XAI).**

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I. INTRODUCTION

AS CONNECTED vehicles proliferate, safety and security risks are rising sharply. With cyberattacks projected to cost the sector $505 billion by 2024 and 84.5% of remote attacks in 2021 [1], scalable IDS solutions are essential to enhance cyber threat detection for connected vehicles. Modern vehicles host many ECUs (e.g., PCM, CCM, BMS, TCU [2]) operating as distributed systems; once an internal network is breached, severe malfunctions and critical safety issues can follow [3], [4]. The attacks may originate from various entry points including sensors, infotainment systems, telematics units, or direct physical interfaces. Recent work reverse-engineering Kia's IVI uncovered major software flaws enabling CAN-bus manipulation across affected models [5]. The evolving threats and the advancement of attack methods highlight the need for adaptive intrusion detection and response systems. Machine learning (ML) solutions show high potential to address this challenge by enabling the IDS to learn from the data representing the state of the vehicle, identify patterns, and adapt to new threats. Thus, a trustworthy IDS must address two aspects: *(1) explainability of decisions* and *(2) achieving generalizability for comprehensive and adaptive detection*. However, the data representing the state of the vehicle under attack is *scarce* as not all potential attack scenarios are known, and even among those known, practical implementation on test vehicles is still limited, due to safety concerns, cost and resource limitations, and ethical considerations. OEMs also limit disclosure of internal network designs [6]. If the IDSs are trained using aggregated or vendor-agnostic statistical features, as shown in [3] and [7], they can work

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and make inference without using proprietary layouts—which can promote generalizability across brands/models and also promote preserving the confidentiality of proprietary data. Following this line, we use CAN telemetry statistics (timing, frequency) instead of raw frames, reducing dependence on OEM-specific content and lowering fleet deployment costs. Guidance from ENISA, NHTSA [8], [9], ISO/SAE 21434 [10], and UNECE WP.29 [11] further supports interoperable, lifecycle-oriented security. By leveraging high-level statistical CAN features (e.g., frequency and timing patterns of message IDs), the proposed IDS is able to detect threats without using proprietary CAN content, aligning with industry and regulatory trends toward unified, scalable security measures.

A. Motivation and Challenges

Although many ML-based IDSs have been proposed, *deployment* in automotive settings still requires investigation [12], [13], [14]. Common models (RF, ANN, SVM) behave as *black-boxes* [15], [16], making false-positive diagnosis and breach forensics difficult—especially in *multi-vendor* contexts [17], [18], [19], [20]. We address this with a *generalized* framework that supports multiple vendors while improving explainability and cross-dataset generalization [21]. Figure 1 contrasts traditional vs. generalized IDS and motivates two core challenges:

- *Challenge I: Scalability and Vendor Specificity.* Standalone, vendor-tailored feature engineering hinders reuse across datasets. Our approach standardizes inputs and applies meta-learning to stay proprietary specifics of individual vendors-agnostic while preserving accuracy.
- *Challenge II: Transparency and Justifiability.* Black-box decisions erode trust. We integrate XAI (SHAP) to provide consistent, dataset-wide and per-window attributions, enabling validation and debugging of IDS behaviour.

A key determinant of ML-based automotive IDS generalizability is deployment. A hybrid design, for example, cloud-based centralized training with on-vehicle inference mitigates the limits of cloud-only or vehicle-only IDS. In this case, the ECUs on the CAN bus host the inference model, while a cybersecurity service provider operates centralized feature engineering and meta-learning–based model building. This enables training with more data and compute, then distributing explainable updates to decentralized onboard models across vendors for tailored yet generalized performance. The model updates follow distributed learning practices [22] and can be delivered periodically or fetched opportunistically by the onboard IDS. There are also a number of automotive standards proposing integration of cryptographic primitives at the network layer to complement IDS functionality [3], [23], [24], [25], [26]. For explainability, SHAP is model-agnostic, efficient, and provides a global view by aggregating per-sample effects, aiding robust tuning and audit across models [17] unlike local-only methods such as LIME [27] or counterfactuals. We therefore pair SHAP with standardized features and meta-learning to couple local fidelity with global interpretability in a generalized IDS. ACHILLES uses centralized training with decentralized deployment of inference model: a vendor-agnostic baseline runs fleet-wide on gateway ECUs. The centrally trained model generalizes across vehicle types and can be deployed directly, giving immediate coverage without per-model data collection. The new patterns learned on any automotive brand are pushed to all vehicles. The timing/frequency features avoid OEM payload disclosure, enabling collaboration. Besides, SHAP on a shared feature space provides consistent, auditable explanations across vehicles. Therefore, the proposed framework scales well when vehicle-specific data are scarce.

B. Summary of Contributions

We develop an ML-based automotive IDS that unifies *explainability* and *generalizability* for centrally supported, multi-vendor fleets. Our key contributions are:

- First, we propose a *centralized training + decentralized execution* IDS for vehicles. We use SHAP to provide model- and feature-level explanations, enabling transparent diagnosis and tuning across vendors.
- Second, we propose a centralized meta-learning process selects effective IDS configurations (DNN/RF and hyperparameters), which are then deployed to on-board units for on-vehicle detection.
- We design standardized, vendor-agnostic CAN features and show, via theory and experiments, that the resulting IDS improves robustness and scalability across heterogeneous vehicles and attack types.

We organize our paper as follows. Section II discusses the related works. In Section III, we describe our proposed framework. In Section IV, we derive the evaluation results. Finally, Section VI concludes the paper.

II. Related Works

We group related studies into automotive IDS, explainable AI (XAI) for IDS, and generalizability, and summarize remaining gaps for a deployable, generalizable, and explainable automotive IDS. *A. Automotive Intrusion Detection:* CAN-based IVNs lack native security, motivating IDS deployment [3]. Prior work spans: *1) Frequency/Timing IDS:* Packet-rate and inter-arrival analysis detect anomalies efficiently [28], [29] but struggle with evolving attacks and cross-vehicle data, *2)Deep Models:* Fine-tuned DNNs on real/synthetic CAN traces improve accuracy [30] yet act as black-boxes for stakeholders, and *3) Hybrid/Advanced:* Signature+anomaly hybrids [31] and attention CNN/LSTM models [32] boost detection, but explanations remain limited. Two gaps persist: (i) *explainability* for safety/audit, and (ii) *multi-vendor generalization*. Our framework targets both via XAI and standardized, vendor-agnostic features across datasets. *B. Explainable AI and IDS:* XAI has been applied to IDS for local/global insight using SHAP/LIME and toolkits like AIX360 [33], [34], [35], [36]. Hybrid rule+ML systems and tree ensembles with SHAP improve interpretability and efficiency [37], [38]. Recent work underscores XAI's value for automotive IDS [17], [39]. *We adopt SHAP to provide consistent local/global attributions for RF/DNN while coupling them with generalized features and meta-learning for model selection. Generalizability and IDS:* Early generate-and-test signature generalization [40] can

Algorithm 1 Meta Learning-Based Model Building Process

Input: Model searching and environment Meta-Data (i.e., $\mathcal{J}$, $f(.)$, $\mathcal{M}_{\text{arc}}$, M, $\mu_{\min}$, $\mu_{\max}$, $\nu_{\min}$, $\nu_{\max}$, $\mathcal{W}$, $\mathcal{W}'$.)
Output: Optimized trained model for IDS
/* Initialization Process */
Initialized learning models as meta-learning agents $j \in J$ and those models with configurations Θ are randomly assigned based on `model_archi_DB`.
/* Competition and Evolution Process */
for Each competition Round $r = 1, \ldots, R$ **do**
  Train $j \in \mathcal{J}$ in parallel or sequential based on the server performance.
  Update the $f(j)$ values of $j \in J$ in Meta Learner's Hierarchical `Q-Tables`.
  Generate random competition pairs $j_k \in \mathcal{J}_{first} \subset \mathcal{J}, \forall j \in \mathcal{J}$.
  for Each $j_k \in \mathcal{J}_{\text{first}}$ **do**
    Determine j'_k and j''_k for each pair $j_k \in \mathcal{J}_{\text{first}}$ based on $f(.)$.
    $j''_k \leftarrow$ **Apply Alg. 2** **`Evolution(`**j'_k, j''_k**`)`**
    Update survivor list $\mathcal{W} \leftarrow j'_k \cup \mathcal{W}$ and non-survivor list $\mathcal{W}' \leftarrow j''_k \cup \mathcal{W}'$.
    `Eliminating Survivors(`$|\mathcal{W}|$**`):`**
    while $|\mathcal{W}| \neq n$ **do**
      Generate elimination stage competition among pairs in $\mathcal{W}$ randomly.
      Update $\mathcal{W}\prime \leftarrow j''_k \cup \mathcal{W}\prime$.
      $\mathcal{W} \leftarrow$**Apply Alg. 2** **`Evolution`** $(\mathcal{W}, \mathcal{W}')$
      Update $\mathcal{W} \leftarrow \mathcal{W} \backslash \mathcal{W}\prime$.
    end while
  end for
end for
/* Model Selection Process */
Choose the model $j^* = \arg\max_{\theta \in \Theta} f(\theta, \mathcal{J})$ which is based-on appropriate metrics (e.g., accuracy, precision).

Algorithm 2 Evolution Process for Model Building

Input: $(j'_k,\ j''_k)$ or $(\mathcal{W}, \mathcal{W}')$
Output: Newly generated model for competition
`Evolution(Input):`
if `rand`(0,1) $> \epsilon$ **then**
  /* Generating evolved model */
  $j''_k \leftarrow$ **Exploit** and construct the evolved model of j'_k with optimal $\theta^* = \arg\max_{\theta \in \Theta} f(\theta, \mathcal{W})$
else
  /* Generating random model */
  $j''_k \leftarrow$ **Explore** randomly and construct learning model with random θ based on `model_archi_DB`.
end if
Return j''_k

miss novel attacks and scale poorly. Standardizing features improves cross-dataset comparability and robustness (e.g., NetFlow formats) [21]; new datasets (IDSAI) further target ML generalization [41]. Foundational work also highlights the difficulty of generalization in neural networks [42]. Our approach advances this line by enforcing vendor-agnostic CAN features and centralized meta-learning, then deploying selected models on-vehicle to improve cross-dataset performance with explainable reasoning.

III. Proposed ACHILLES Framework for Automotive IDS

The proposed intrusion detection framework has three main parts: a) data preprocessing, b) IDS model building, and c) automotive IDS explanation and generalization approach.

A. Data Preprocessing

Most CAN bus messages arrive at fixed frequencies, and attackers often inject anomalous frames that disrupt these patterns [28], [29]. To detect such deviations, we adopt a time- and frequency-based preprocessing strategy, as shown in the *Data Preprocessing* block of Fig. 2. First, the raw CAN logs are parsed by a microcontroller and then cleaning, balancing, and scaling is conducted to produce aggregated features. Using four public datasets (described in detail in Section IV-A), we apply a sliding window approach [43] with a window size of $n = 10$, which has been shown to be effective in capturing timing and frequency trends. Each window of n consecutive messages, shifted by one message at a time, yields overlapping windows that preserve local context and increase data volume. This uniform method is applied across all datasets. Each window produces features in two categories. In Table I, first, standard CAN fields include timestamps (`t`), data bytes (`D0–D7`), and message IDs (`ID_n`). Second, we compute a set of custom and generalized metrics such as inter-arrival times (`dt_n`, `dt_ID_n`), frequency counts of the most/least frequent message IDs (`cmafid`, `cmifid`) and payloads (`cmafpl`, `cmifpl`), counts of unique IDs/payloads (`uidc`, `nuidc`, `uplcw`, `nuplc`), and the mean inter-frame gap (`wmdt`). These features are designed to highlight anomalies caused by message injection or manipulation. Each window is labeled as *attack* or *normal* based on the presence of malicious messages, yielding a supervised learning dataset. Compared to [43], our method incorporates new, tailored features with higher importance to test model generalizability and explainability. By standardizing feature construction and focusing on temporal-frequency signals, the framework supports robust IDS model development and facilitates XAI techniques (e.g., SHAP) to trace model decisions to CAN-level indication for malicious behavior.

B. IDS Model Building

The *IDS Model Building* block of the ACHILLES framework in Fig. 2 involves finding the appropriate ML model configurations for a given set of the model architecture $\mathcal{M}_{arch}$. Therefore, the proposed meta-learning approach searches for the best configurations θ^* for DNN and RF models. The model building stage is comprised of two algorithms, model-building (i.e., Alg. 1) and model evolution process (i.e., Alg. 2).

1) Model-Building Algorithm: In Alg. 1, the meta-learning approach consists of three main steps: *initialization*, *competition and evolution*, and *model selection*. The meta-learning

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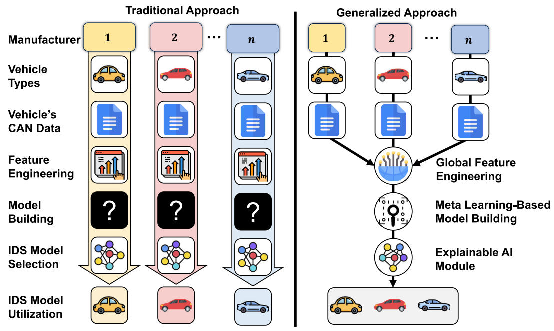

Fig. 1. Comparison between the traditional and generalized approach for automotive IDS development.

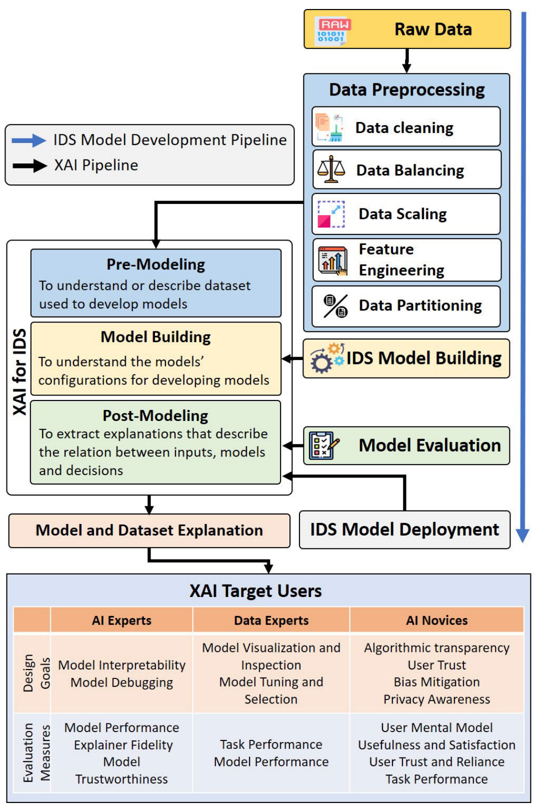

Fig. 2. Proposed ACHILLES framework with explainablility phases for IDS construction and understanding IDS decisions for generalizability.

algorithm is set up and initialized during the initialization step with default parameters (i.e., model's type, model's configuration, initialized model's weights with the help of normal distribution) from a given model architecture and configuration database, `model_archi_DB` (lines $1-4$, in Alg. 1). The initialized learning models are considered as a set of meta-agents $\mathcal{J}$, and the performance of the agents is evaluated with a fitness/reward function $f(.)$. In the case of DNN, the objective of Alg. 1 is to find the model-specific meta-data, such as minimum (μ_{min}) and maximum (μ_{max}) number of hidden layers, minimum (ν_{min}) and maximum (ν_{max}) number of neurons in the hidden layers (line 1, in Alg. 1). During the competition and evolution processes, the survivor/winner and non-survivor/rejected model pairs are stored in the sets $\mathcal{W}$ and $\mathcal{W}'$, respectively. The competition process involves running multiple iterations of the first and elimination stage competitions and evolution, using different configurations for each iteration (lines $5-22$, in Alg. 1). Finally, in the model selection step, the algorithm selects the model configuration that performed the best during the competition step (lines $23-24$, in Alg. 1). The number of competition rounds R is finite, where at each round r, the competition and evolution processes involve finding the optimized trained model and configuration for the automotive IDS (line 6, in Alg. 1). Therefore, the meta-learning agents in $\mathcal{J}$ are trained in parallel, where the performance of each $j \in \mathcal{J}$ is evaluated through the fitness function $f(j)$ (line 7, in Alg. 1). The performance value is then updated during each r and stored in the meta-learners' hierarchical Q-table (line 8, in Alg. 1). Through random exploration of $\mathcal{J}$, the random competition pairs j_k are generated and stored in a $\mathcal{J}_{first} \subset \mathcal{J}$ (line 9, in Alg. 1). Based on model fitness value $f(.)$, the survivor model j_k' and non-survivor model j_k'' are determined for each competition pair $j_k \in \mathcal{J}_{first}$ (lines $10-11$, in Alg. 1). Using j_k' and j_k'', the evolution process is applied through Alg. 2 within the competition round r of model building (line 12, in Alg. 1).

2) Model Evolution Process: In Alg. 2, the exploration and exploitation operations within the evolution process are performed to generate either an evolved model or a random one (lines $3-10$, in Alg. 2). If the random value between $[0, 1]$ is greater than a probability threshold ϵ, the exploitation operation generated evolved models in j_k'' with the best known searching path based on the meta-learner's hierarchical Q-table and constructs the evolved model of j_k' with optimal configuration θ^* (lines $4-6$, in Alg. 2). Otherwise, the model evolution is done through random exploration operation where the searching path is explored randomly to construct and update the learning models j_k'' with random configuration θ based on initially given model architecture (lines $7-9$, in Alg. 2). Once the model configurations are determined for the model pairs in $\mathcal{J}_{first}$, the survivor and non-survivor lists, $\mathcal{W}$ and $\mathcal{W}'$ are updated (line 13, in Alg. 1). The elimination from the survivor list $\mathcal{W}$ is done to achieve a fixed number of models (n) from $|\mathcal{W}|$ and their configurations (lines $14-20$, in Alg. 1). In the elimination stage, the non-survivor list $\mathcal{W}'$ is updated by determining the eliminated winner j_k'' from pairs in $\mathcal{W}$ based on $f(.)$ (lines $16-17$, in Alg. 1). At this point, using $\mathcal{W}$ and $\mathcal{W}'$, the evolution process is applied again through Alg. 2 to find the model configurations and the final survival list $\mathcal{W}$ (lines $17-18$, in Alg. 1). With fixed $\mathcal{W}$, the model selection process results in a suitable configured model with appropriate model performance evaluation metrics such as model accuracy and precision (lines $23-24$, in Alg. 1). A detailed discussion of the evaluation metrics is provided later in Section IV-B. The configured trained model from the *IDS model building* phase is deployed at the vehicles' OBU for on-vehicle decision-making at the *IDS Model Deployment* phase in Fig. 2.

C. Automotive IDS Explanation and Generalization Approach

A typical IDS model development involves preprocessing raw data that undergoes several steps, such as data cleaning, data balancing, data scaling, feature engineering, and data partitioning. The IDS model is often hard to explain and is evaluated against standard evaluation metrics. In this proposal, as shown in Fig. 2, we add the *Model and Dataset Explanation* block for IDS construction, which supports the IDS model building by going through stages of pre-modeling (i.e., dataset explanation), model building (i.e., model's configuration explanation), and post-modeling (i.e., explanation of relations between inputs, models and decisions). This approach allows a more reliable evaluation of the ML models that are otherwise blackbox systems. The model and dataset explanation can be helpful for AI experts focusing on model interpretability to achieve performance explanation and trust. Also, data experts interested in model visualization and tuning can better evaluate task and model performance. AI novices (including traditional cybersecurity analysts using AI techniques) can benefit from algorithmic transparency, trust in the detection systems, bias mitigation, and privacy awareness for better task performance. In addition, the traditional ML models generate specific predictions after receiving some features as input. However, following the ML-based IDS model training, the user can have questions such as why the model predicts the input as attack or non-attack, when the model succeeds in detecting attacks, and when it fails, when it can be trusted, and how can an error be corrected. Therefore, the model's explainability significantly impacts ML applications especially in safety-critical scenarios such as automotive systems. Besides debugging, directing feature engineering, data collection, decision-making, and building trust, a model's explainability can also support creating a generalized model that will fit well to multiple datasets from multiple vendors by providing the contributions of the involved features. The models can be explained using a variety of methods such as LIME, SHAP, counterfactual explanation [35]. This paper considers SHAP (SHapley Additive exPlanations) values as the method can be essentially leveraged to compute the contribution of each feature to the prediction, which aids the process of coming up with a set of generalized features. Moreover, SHAP allows a simpler evaluation of Shapley values with fewer coalition samples than the other explainable AI approaches. In this work, the contribution of each attack prediction can be measured using the input features by utilizing SHAP values in the ML model explanation under the automotive IDS framework. Some advantages of using SHAP values over other SOTA techniques for automotive IDS include: 1) the identification of additive CAN bus data feature importance measures contributing to an attack prediction for vehicles, 2) guarantee of a unique attack detection outcome with a set of relevant properties, 3) better compliance to the understanding of cybersecurity and automotive expert intuitions, and 4) efficient computational complexity in terms of achieving fast convergence and discriminating among prediction outcomes for automotive IDS. In a typical linear model, the prediction for a single data point x can be represented as

$$\hat{f}(x) = \xi_0 + \xi_1 x_1 + \ldots + \xi_J x_J, 1 \leq j \leq J, \tag{1}$$

where each x_j is represented as feature value, ξ_0 denotes the initial weight, and ξ_j denotes the weight corresponding to a feature j. Using (1), the contribution Φ_j of the j^{th} feature on the prediction $\hat{f}(x)$ is calculated as [44],

$$\Phi_j(\hat{f}) = \xi_j x_j - \xi_j \mathbb{E}(X_j), \tag{2}$$

where $\mathbb{E}(\xi_j X_j)$ is the mean effect estimate for feature j. In (2), the feature contribution, $\Phi_j(\hat{f})$, is measured by subtracting the average effect from the feature effect. By using (2), the sum of all the feature contributions for one instance is defined as,

$$\begin{aligned}\sum_{j=1}^{J} \Phi_j(\hat{f}) &= \sum_{j=1}^{J} (\xi_j x_j - \mathbb{E}(\xi_j X_j)) \\ &= \left(\xi_0 + \sum_{j=1}^{J} \xi_j x_j\right) - \left(\xi_0 + \sum_{j=1}^{J} \mathbb{E}(\xi_j X_j)\right) \\ &= \hat{f}(x) - \mathbb{E}(\hat{f}(X)).\end{aligned} \tag{3}$$

To capture the marginal contribution of a feature j in a prediction outcome, we apply the Shapley value as [45],

$$\begin{aligned}\Phi_j(v) = \sum_{\mathbb{S} \subseteq \{1,\cdots,J\} \setminus \{j\}} \frac{|\mathbb{S}|!(J - |\mathbb{S}| - 1)!}{J!} \\ (v(\mathbb{S} \cup \{j\}) - v(\mathbb{S})).\end{aligned} \tag{4}$$

Here $\mathbb{S}$ is a subset of the features used in the model that excludes the feature j in the prediction where x represents the feature values of the instance to be explained. In other words, the Shapley value of a feature indicates its contribution to the prediction outcome that is weighted and summed over all possible feature value combinations where J is the number of features. Using (4), the feature contributions of a single prediction outcome for any ML model can be defined via a value function v of feature set $\mathbb{S}$. Therefore, $v_x(\mathbb{S})$ denotes a prediction for feature values that are marginalized over the features that are not included in set $\mathbb{S}$ and calculated as,

$$v_x(\mathbb{S}) = \int \hat{f}(x_1, \ldots, x_J) d\mathbb{Q}_{x \notin \mathbb{S}} - \mathbb{E}_X(\hat{f}(X)), \tag{5}$$

where $\mathbb{Q}$ is a set of features that does not include the feature instance[1] $x \in \mathbb{S}$.

When these datasets[2] are sent to the cloud, each dataset goes through the preprocessing phases before being trained by the machine learning detection models. Our proposal considers preprocessing datasets with standard feature formats, as discussed in Section III-A, to facilitate dataset generalization. To verify the generalization of the feature formats, we employ the training with one vehicle dataset and validate with another. As a result, a set of possible test combinations are generated whose results are then compared. The generalization of the test combination results is further evaluated with the above-described XAI technique by extracting the associated feature importance in each test case. It is worth noting that our proposed framework is fundamentally designed to be modular

[1]Equation (5) can be extended to perform the prediction, $v_x(\mathbb{S})$, for more than one features that are not included in set $\mathbb{S}$ by applying multiple integration operation.

[2]Different publicly available CAN datasets are originally sourced from various vehicle models. While they are typically anonymized or open-sourced via research institutions (not directly by OEMs), these publicly available datasets are from vehicles with heterogeneous specifications.

and scalable, leveraging state-of-the-art ML technologies. It can be seamlessly integrated into any ML-based automotive IDS that utilizes current leading-edge machine learning models. Specifically, our framework employs a standardized feature format based on CAN bus data, which is ubiquitous in automotive systems. This standardization ensures that as new ML models emerge, they can be easily incorporated by simply adjusting the input features to fit this established format. Furthermore, the meta-learning component of our framework is designed to optimize hyperparameters dynamically, which is essential for adapting to new ML models that might have different requirements or capabilities. This approach allows for continuous learning and adaptation without the need for extensive manual reconfiguration. To assess and ensure the generalizability and explainability of these new models within our framework, we employ SHAP which provides a consistent and robust method to evaluate the impact of features in ML models. These properties of SHAP make it an invaluable tool for maintaining transparency and understanding in IDS applications as model technologies evolve while the model-agnostic nature of SHAP allows it to be applied consistently across different types of machine learning models.

D. Complexity Analysis

Computing SHAP on large datasets and complex models can be costly; we therefore summarise the dominant terms for our IDS. For a DNN with H hidden layers and N neurons per layer, the SHAP computation scales as $\mathcal{O}(H\,N\,M)$ for M samples—linear in M but increasing with depth/width [46], [47]. For a Random Forest with $|S|$ features and T trees, training is $\mathcal{O}(M\,|S|\,\log|S|)$ and SHAP is $\mathcal{O}(M\,|S|\,T)$ [48]. These complexities are practical on our target platforms and align with prior scalability studies in high-dimensional IDS settings [39]. In short, while cost grows with data size and model capacity, the DNN/RF choices and SHAP-based explanations remain tractable for automotive IDS and provide transparent, defensible decisions.

IV. Performance Evaluation

A. Datasets

We evaluate on four public in-vehicle CAN datasets:

- *Survival Analysis Dataset (SAD)* [49]: Normal driving plus three *real* injection attacks (DoS/flooding, fuzzing, malfunction) from Hyundai Sonata, Kia Soul, and Chevrolet Spark. Attacks last 25–100 s with 5 s bursts. Limitation: few, relatively simple attack types.
- *Real ORNL Automotive Dynamometer Dataset (ORNL)* [50]: Physically verified CAN attacks (e.g., fuzzing, fabrication, masquerade) and two advanced non-fabrication attacks. Logs captured on-road and on a chassis dynamometer; vehicle identity (make/model/region) is anonymized via metadata redaction.
- *CAN intrusion detection dataset (CAN-IDS)* [51]: Normal traffic plus synthetic DoS, fuzzing, and impersonation (masquerade) with labeled records (Timestamp, ID, DLC, DATA[0–7]). Masquerade relies on spoofed remote-frame behavior; utility may depend on IDS use of that protocol aspect; injection timing details are coarse.
- *Car Hacking: Attack & Defense Challenge 2020 Dataset (CAR-HACK)* [52]: Normal and four *real* attack types (DoS/flooding, fuzzing, replay, spoofing) across preliminary/submission/final rounds (the last stationary for safety). Rich captures with many instances per attack; fields include Timestamp, Arbitration_ID, DLC, Data, Class, SubClass.

We chose SAD, ROAD, CAN-IDS, and CAR-HACK for (i) *attack diversity* (fuzzing, replay, fabrication, masquerade, etc.), (ii) *realistic vs. simulated coverage* (ROAD includes both physically verified and simulated attacks), and (iii) *community adoption* for benchmarking. Cross-dataset validation mitigates overfitting to any single source, and our explainable meta-learning aims to detect anomalies that generalize beyond dataset-specific artifacts.

TABLE I

Summary of Features

Notation	Description
`t`	Existing timestamp of the collected CAN bus data
`D0-D7`	Existing data where `D0` is the first byte, and `D7` is the last byte of the payload from CAN bus data
`ID_n`	Existing `ID` of the `n` message frames (e.g., if window size is 10, `n` ∈ [1..10], such as `ID_0` to `ID_10`)
`pl_n`	Payload of the `n` message frames (e.g., if window size is 10, `n` ∈ [1..10], such as `payload_0` to `payload_10`)
`dt_n`	Time difference between timestamps of 10 message frames (e.g., if window size is 10, `n` ∈ [1..10], such as `dt_0` to `dt_10`)
`dt_ID_n`	Time difference between each individual CAN ID of `10` frames (e.g., if window size is 10, `n` = [1..10], such as `dt_ID_0` to `dt_ID_10`)
`cmafid`	Number of occurrences of the most frequent CAN ID per window.
`cmifid`	Number of times the ID with minimum frequency has occurred per window.
`cmafpl`	Number of times the payload with maximum frequency has occurred per window.
`cmifP`	Number of times the payload with minimum frequency has occurred per window.
`uidc`	Number of IDs that have uniquely occurred per window
`nuidc`	Number of IDs that have more than one occurrence per window.
`uplcw`	Number of payloads that have uniquely occurred per window.
`nuplc`	Number of payloads having more than one occurrence per window.
`wmdt`	Mean time difference between the CAN- frames in the sliding window.

B. Performance Analysis of the Proposed Approach

The performance of the proposed framework was evaluated on a machine with 128 GB RAM, an Intel Core i7-6850K CPU @ 3.60 GHz (6 cores, 12 threads), and two NVIDIA TITAN X GPUs (Pascal architecture, GP102, 12 GB VRAM each). The test combinations generated for generalization evaluation are tested with a baseline neural network [53], achieving high performance in automotive IDS evaluation compared with meta-learning-based DNN and meta-learning-based RF. The choice of the selected models also allows evaluating both the shallow learning and deep learning method. The hyperparameters are tuned with the adopted meta-learning approach. The meta-learning-based DNN model configuration

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TABLE II
BASELINE MODEL [53] GENERALIZATION ANALYSIS WITHOUT META-LEARNING PERFORMANCE

Dataset (Train-Test)	Time	Accuracy	Precision	Recall	F1 score	Cohens kappa	ROC AUC
ORNL-ORNL	4.23	0.9980	0.9970	**0.9990**	0.9980	0.9960	0.9980
SAD-SAD	4.25	0.9865	0.9928	0.9798	0.9863	0.9730	0.9864
CAN-IDS-CAN-IDS	4.41	0.8930	0.9738	0.8057	0.8818	0.7856	0.8922
CAR-HACK-CAR-HACK	4.41	0.9942	0.9915	0.9970	0.9942	0.9885	0.9943
$\text{ORNL}_{\text{proposed}}$-$\text{ORNL}_{\text{proposed}}$	4.26	0.9800	0.9800	0.9800	**1.0000**	**1.0000**	**1.0000**
$\text{SAD}_{\text{proposed}}$-$\text{SAD}_{\text{proposed}}$	4.42	**0.9995**	0.9910	**0.9990**	0.9995	0.9990	0.9995
$\text{CAN-IDS}_{\text{proposed}}$-$\text{CAN-IDS}_{\text{proposed}}$	4.44	0.8069	0.8199	0.7898	0.8046	0.6138	0.8070
$\text{CAR-HACK}_{\text{proposed}}$-$\text{CAR-HACK}_{\text{proposed}}$	4.50	0.9987	**0.9990**	0.9985	0.9988	0.9975	0.9988
ORNL-SAD	4.11	0.4835	0.3488	0.0495	0.0866	-0.0415	0.4794
SAD-ORNL	4.28	0.5480	0.5671	0.3690	0.4471	0.0929	0.5463
CAN-IDS-CAR-HACK	4.40	0.6185	0.6508	0.4957	0.5628	0.2352	0.6173
CAR-HACK-CAN-IDS	4.43	0.4785	0.4695	0.4079	0.4365	-0.0444	0.4778
$\text{ORNL}_{\text{proposed}}$-$\text{SAD}_{\text{proposed}}$	4.32	0.8049	0.8449	0.7500	0.7946	0.6100	0.8052
$\text{SAD}_{\text{proposed}}$-$\text{ORNL}_{\text{proposed}}$	4.38	0.7471	0.8084	0.6521	0.7219	0.4948	0.7477
$\text{CAN-IDS}_{\text{proposed}}$-$\text{CAR-HACK}_{\text{proposed}}$	4.55	0.7763	0.7357	0.8673	0.7961	0.5521	0.7757
$\text{CAR-HACK}_{\text{proposed}}$-$\text{CAN-IDS}_{\text{proposed}}$	4.47	0.6540	0.6569	0.6546	0.6557	0.3080	0.6540
Identical train-test Dataset (Average)	4.33	0.9679	0.9888	0.9454	0.9651	0.9358	0.9677
Proposed identical train-test Dataset (Average)	4.41	0.9463	0.9475	0.9418	0.9507	0.9026	0.9513
Cross train-test Dataset (Average)	4.31	0.5321	0.5091	0.3305	0.3833	0.0606	0.5302
Proposed cross train-test Dataset (Average)	4.43	0.7456	0.7615	0.731	0.7421	0.4912	0.7457

TABLE III
PROPOSED DNN MODEL GENERALIZATION ANALYSIS WITH META-LEARNING PERFORMANCE

Dataset (Train-Test)	Time (s)	Accuracy	Precision	Recall	F1 score	Cohens kappa	ROC AUC
ORNL-ORNL	4.50	0.998	0.997	0.999	0.998	0.996	0.998
SAD-SAD	4.54	0.995	0.996	0.994	0.995	0.990	0.995
CAN-IDS-CAN-IDS	4.65	0.949	0.982	0.914	0.947	0.898	0.949
CAR-HACK-CAR-HACK	4.69	0.999	0.9995	0.9985	0.999	0.998	0.999
$\text{ORNL}_{\text{proposed}}$-$\text{ORNL}_{\text{proposed}}$	4.54	**0.9997**	**1.0000**	**0.9995**	**0.9998**	**0.9995**	**0.9998**
$\text{SAD}_{\text{proposed}}$-$\text{SAD}_{\text{proposed}}$	4.68	**0.9997**	0.9998	**0.9995**	**0.9998**	**0.9995**	**0.9998**
$\text{CAN-IDS}_{\text{proposed}}$-$\text{CAN-IDS}_{\text{proposed}}$	4.63	0.893	0.920	0.863	0.891	0.786	0.893
$\text{CAR-HACK}_{\text{proposed}}$-$\text{CAR-HACK}_{\text{proposed}}$	5.41	**0.9997**	0.999	**0.9995**	**0.9998**	**0.9995**	**0.9998**
ORNL-SAD	4.51	0.491	0.422	0.077	0.130	-0.027	0.487
SAD-ORNL	4.51	0.522	0.522	0.360	0.427	0.040	0.520
CAN-IDS-CAR-HACK	4.63	0.692	0.647	0.833	0.728	0.386	0.693
CAR-HACK-CAN-IDS	4.65	0.482	0.478	0.512	0.494	-0.036	0.482
$\text{ORNL}_{\text{proposed}}$-$\text{SAD}_{\text{proposed}}$	4.54	0.708	0.849	0.510	0.637	0.417	0.709
$\text{SAD}_{\text{proposed}}$-$\text{ORNL}_{\text{proposed}}$	4.60	0.719	0.733	0.695	0.713	0.438	0.719
$\text{CAN-IDS}_{\text{proposed}}$-$\text{CAR-HACK}_{\text{proposed}}$	4.70	0.793	0.716	0.975	0.826	0.584	0.791
$\text{CAR-HACK}_{\text{proposed}}$-$\text{CAN-IDS}_{\text{proposed}}$	4.74	0.675	0.663	0.720	0.690	0.349	0.674
Identical train-test Dataset (Average)	4.60	0.9853	0.9936	0.9764	0.9848	0.9705	0.9853
Proposed identical train-test Dataset (Average)	4.82	0.9730	0.9797	0.9654	0.9726	0.9461	0.9731
Cross train-test Dataset (Average)	4.58	0.5468	0.5173	0.4455	0.4448	0.0908	0.5455
Proposed cross train-test Dataset (Average)	4.65	0.7238	0.7403	0.725	0.7165	0.447	0.7233

among the constructed DNN configuration results in a model having an input layer with 256 neurons, dense layers with 128 and 64 neurons, and a final output layer with two neurons. For random forest-based automotive IDS, the meta-learning-based model configuration consists of 100 estimators (i.e., RF). We tested the models with a training size of 0.2, i.e., 20% of the data, and a test size of 0.8, i.e., 80% of the data, as it is an effective approach to testing the generalization of a model, especially when a smaller amount of data is available [54]. We consider the original datasets, as ORNL, SAD, CAN-IDS, CAR-HACK, respectively. $\text{ORNL}_{\text{proposed}}$, $\text{SAD}_{\text{proposed}}$, $\text{CAN-IDS}_{\text{proposed}}$,and $\text{CAR-HACK}_{\text{proposed}}$are the datasets containing the new features discussed in Table I.

Table II presents the baseline model's [53] performance comparison without the proposed meta-learning model search between the identical and cross-dataset train and test data scenarios. The analysis includes metrics like accuracy, precision, recall, F1 score, Cohen's kappa, ROC AUC, and average inference time per model (in seconds) measured over the testing dataset. Specifically, we calculate on-average how long each IDS model requires to process a given CAN frame or batch of frames during inference. This metric provides insight into each model's computational efficiency under realistic conditions, complementing the accuracy-related performance metrics (accuracy, precision, recall, F1, Cohen's kappa, and ROC AUC). The performance is compared across different training and testing datasets, including ORNL, SAD, CAN-IDS, and CAR-HACK. In the case of identical dataset (ORNL, SAD, CAN-IDS, and CAR-HACK) scenarios, the average performance in terms of accuracy is approximately 0.9679. This performance indicates a high level of overall effectiveness when the baseline model is trained and tested on the same dataset. More specifically, the identical ORNL train and test data scenario exhibits the highest accuracy of 0.9980. Like ORNL, the CAR-HACK dataset in baseline model displayed significant performance gain (i.e., the accuracy of 0.9942), indicating robust learning from this dataset. The performance with proposed ORNL and SAD dataset enhancements with the generalized feature depicts improvements in accuracy (0.9800 and 0.9995), showcasing the effectiveness of the proposed dataset enhancements. Similarly, considerable improvements are visible in the proposed CAR-HACK scenario, where

TABLE IV

PROPOSED RF MODEL GENERALIZATION ANALYSIS WITH META-LEARNING PERFORMANCE

Dataset (Train-Test)	**Time (s)**	**Accuracy**	**Precision**	**Recall**	**F1 score**	**Cohens kappa**	**ROC AUC**
ORNL-ORNL	0.04	0.9982	0.9965	**1.0000**	0.9982	0.9965	0.9983
SAD-SAD	0.04	0.9990	0.9999	0.9980	0.9990	0.9980	0.9990
CAN-IDS-CAN-IDS	0.05	0.9782	0.9897	0.9662	0.9778	0.9565	0.9781
CAR-HACK-CAR-HACK	0.04	0.9999	0.9999	0.9999	0.9999	1.0000	0.9999
$\text{ORNL}_{\text{proposed}}$-$\text{ORNL}_{\text{proposed}}$	0.03	0.9997	0.9999	0.9995	0.9998	0.9995	0.9998
$\text{SAD}_{\text{proposed}}$-$\text{SAD}_{\text{proposed}}$	0.03	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
$\text{CAN-IDS}_{\text{proposed}}$-$\text{CAN-IDS}_{\text{proposed}}$	0.06	0.9477	0.9999	0.8961	0.9452	0.8955	0.9481
$\text{CAR-HACK}_{\text{proposed}}$-$\text{CAR-HACK}_{\text{proposed}}$	0.04	0.9997	0.9999	0.9995	0.9998	0.9995	0.9998
ORNL-SAD	0.04	0.5048	0.0000	0.0000	0.0000	0.0000	0.5000
SAD-ORNL	0.04	0.5807	0.5773	0.5729	0.5751	0.1614	0.5807
CAN-IDS-CAR-HACK	0.04	0.7093	0.6306	0.9970	0.7725	0.4216	0.7120
CAR-HACK-CAN-IDS	0.03	0.4467	0.4393	0.4235	0.4313	-0.1070	0.4465
$\text{ORNL}_{\text{proposed}}$-$\text{SAD}_{\text{proposed}}$	0.04	0.7991	0.9992	0.6014	0.7509	0.5993	0.8004
$\text{SAD}_{\text{proposed}}$-$\text{ORNL}_{\text{proposed}}$	0.04	0.7531	0.8684	0.6004	0.7100	0.5071	0.7541
$\text{CAN-IDS}_{\text{proposed}}$-$\text{CAR-HACK}_{\text{proposed}}$	0.04	0.5154	0.5095	0.9999	0.6751	0.0243	0.5121
$\text{CAR-HACK}_{\text{proposed}}$-$\text{CAN-IDS}_{\text{proposed}}$	0.03	0.7023	0.7236	0.6610	0.6909	0.4049	0.7026
Identical train-test Dataset (Average)	0.04	0.9938	0.9965	0.9910	0.9937	0.9878	0.9938
Proposed identical train-test Dataset (Average)	0.04	0.9868	0.9999	0.9738	0.9862	0.9736	0.9869
Cross train-test Dataset (Average)	0.04	0.5604	0.4118	0.4984	0.4447	0.119	0.5598
Proposed cross train-test Dataset (Average)	0.04	0.6925	0.7752	0.7157	0.7067	0.3839	0.6923

accuracy increased to 0.9987. There was a slight degradation in performance for proposed CAN-IDS dataset (0.8930) compared to original CAN-IDS data (0.8069) in identical train-test setup which could be caused by baseline feature set becoming highly tuned. However, proposed CAN-IDS showed better generalizability in cross train-test (0.7763 and 0.6540) compared to original CAR-HACK cross train-test (0.6185 and 0.4785). In the case of efficiency and response time, the model maintains a similar response time across both original and proposed features in identical and cross train-test scenarios, with average times within the range of 4.31 to 4.43 seconds, indicating its efficiency in processing. *The baseline model consistently performed well in identical scenarios, indicating a strong fit to specific dataset characteristics, the generalization challenges remain for the baseline method without the meta-learning. In particular, the decreases in performance in cross-dataset scenarios highlight the baseline model's difficulties in generalizing its learning to new, unseen datasets.*

Table III presents a comparative performance analysis of the proposed DNN model with meta-learning, focusing on proposed meta-learning and proposed generalizable feature capabilities for model inference. The highest performance across all metrics is observed when training and testing are done on the same dataset and especially with proposed feature formatted datasets. For instance, in the proposed ORNL and SAD datasets, almost all metrics are close to 1 (99.97% accuracy, 100% precision, 99.95% recall, etc.). This trend is also seen in the proposed CAR-HACK dataset (99.97%) but there is a similar drop in the proposed CAN-IDS dataset (89.3%) compared to original CAN-IDS dataset (94.9%) in the identical train-test scenario but the performance of the proposed DNN model in both cases is better than the baseline model in Table II. Moreover. proposed CAN-IDS performs better (79.3% and 67.5%) than original CAN-IDS (69.2% and 48.2%) in the cross train-test scenario. The cross train-test dataset performance degradation is also visible in the proposed DNN model with original datasets and a performance gain with proposed datasets. For example, when the origical ORNL dataset is tested on the original SAD datset, the accuracy drops to 49.1%, and similarly, the original SAD dataset tested on original ORNL dataset shows a 52.2% accuracy. This indicates a challenge in generalizing across different datasets, even though the proposed models generally show significant improvement. For example, comparing the original CAN-IDS dataset tested on itself (94.9% accuracy) with the proposed CAN-IDS dataset (89.3% accuracy), there's a slight decrease in performance. However, when tested on CAR-HACK, the proposed dataset shows a substantial improvement (from 69.2% to 79.3% accuracy), indicating better generalization. This indicates a 10.1% increase in accuracy, showcasing the effectiveness of the proposed model in adapting to different data types. High Cohen's kappa and ROC AUC values in the proposed datasets (especially in homogeneous training and testing scenarios) indicate strong reliability and the ability to distinguish between classes effectively. For instance, the proposed CAR-HACK dataset has a Cohen's kappa of 0.9995 and a ROC AUC of 0.9998 when tested on its own. In the case of efficiency and response time, the model maintains a reasonable response time similar to the baseline model across scenarios, with average times ranging between 4.58 to 4.82 seconds, indicating its consistent efficiency in processing. *Overall, Meta-learned DNNs outperform the baseline in three of four averaged settings which are, identical train–test, proposed-identical train–test, and cross train–test. The results demonstrate the effectiveness of the meta-learning model-building process and the proposed generalized feature for cross-dataset generalization. These gains indicate a more robust, adaptable DNN for complex datasets.*

Table IV presents an extensive evaluation of the Proposed RF model enhanced with meta-learning. In the case of identical train-test scenarios, the original ORNL data achieves top-tier performance with an accuracy of 0.9982, precision of 0.9965, and recall at 1.0000. Similarly, significant performance gains in terms of accuracy are observed in SAD, CAN-IDS, and CAR-HACK (i.e., 0.9990, 0.9782, and 0.9999). The proposed datasets performance for ORNL and SAD are also significantly high (i.e., 0.9997 and 1.0000) compared

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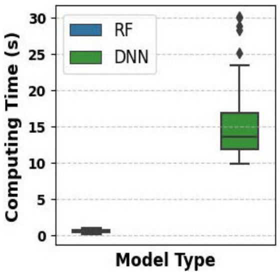
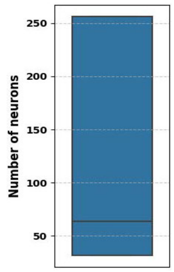
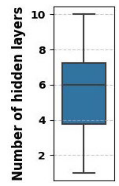


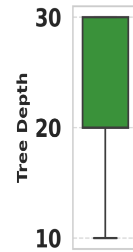
(a) Computation time between DNN and RF models.



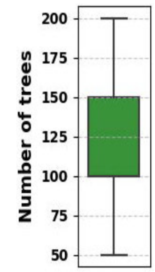
(b) DNN neurones.



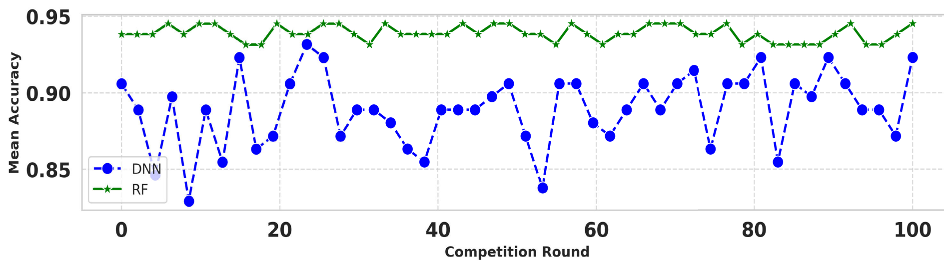
(c) DNN hidden layers.



(d) RF depth.



(e) DT in RF.



(f) Validation accuracy for each competition round during model building process.

Fig. 3. Performance evaluation of the deep neural network (DNN) and random forest (RF) model building stage of the proposed ACHILLES framework.

to the original datasets in the proposed RF model and both original and proposed datasets in the baseline model. There is a slight performance degradation in the CAN-IDS dataset as observed before in both baseline and proposed DNN model. The performance in the proposed CAR-HACK (0.9997) remains relatively same with very slight change as original CAR-HACK (0.9999). In the case of cross-dataset scenarios, the model faces the expected challenge with a drop in performance. For example, a drop in original ORNL-SAD accuracy to 0.5048 and zero precision and recall, indicating difficulty in applying learnings from ORNL to SAD. However, the original SAD-ORNL case displays better adaptability with an accuracy of 0.5807. In the case of CAN-IDS - CAR-HACK, the accuracy is moderate at 0.7093 and CAR-HACK - CAN-IDS, there is a significant drop to 0.4467. However, all these four results are higher compared to the base line model (i.e., 0.4835, 0.5480, 0.6185, and 0.4785). In the case of the proposed datasets on the proposed RF model, there is also significant improvement in all four cases (i.e., 0.7991, 0.7531, 0.5154, and 0.7023). The proposed SAD to proposed ORNL (0.7991) and the proposed CAR-HACK to proposed CAN-IDS (0.7023) in the proposed RF model is also higher than the base line model corresponding cross train-test scenario (i.e., 0.7471 and 0.6540, respectively). In the case of efficiency and response time, the model not only demonstrates high accuracy but also maintains a rapid response time across scenarios, with times ranging from 0.03 to 0.04 seconds, indicating its efficiency in processing. *Overall, the proposed RF models with meta-learning show performance gain on three out of four average dataset tests, i.e., identical train-test dataset (average), proposed identical train-test dataset (average), and cross train-test dataset (average), compared to the baseline model further confirming the effectiveness of the meta-learning-based model-building process and generalization capability of the proposed generalized features.*

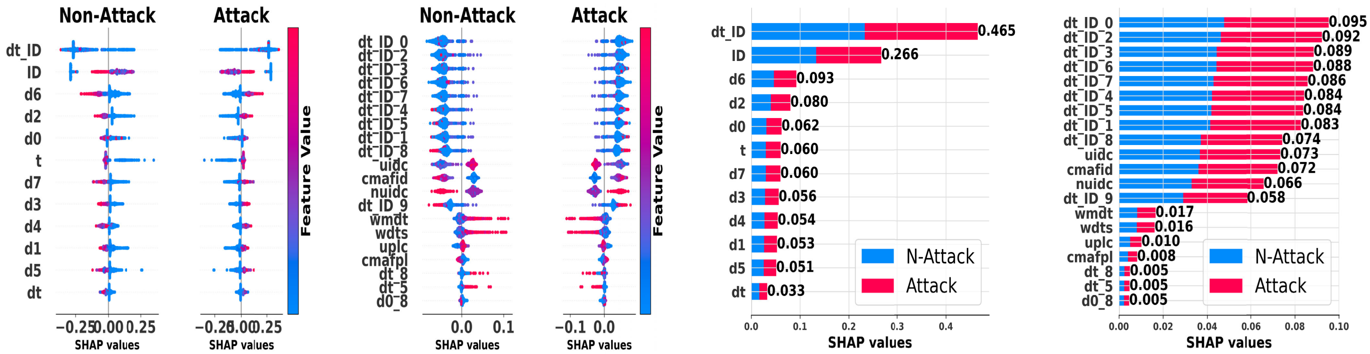
A comparative performance analysis among the baseline model without meta-learning, the proposed DNN with meta-learning, and the proposed RF models with meta-learning are revealed through the detailed Table II, Table III and Table IV, respectively. It is clear from all the three tables that, in general, the highest performance gain (i.e., the numbers in bold) is achieved with the proposed feature formatted datasets that are trained and tested with the identical datasets. Moreover, the average accuracy of proposed cross train-test datasets is higher (0.7456 with Baseline, 0.7238 with DNN using meta-learning, 0.6925 with RF using meta-learning) than the average accuracy of original cross train-test datasets (0.5321 with Baseline, 0.5468 with DNN using meta-learning, 0.5604 with RF using meta-learning). Regarding cross-dataset generalization, the accuracy of the ORNL for train data and SAD for test data scenario is 0.4835 in Table II. On the other hand, in Table III, the proposed DNN model with meta-learning with proposed ORNL train data and proposed SAD test data demonstrates an accuracy of 0.708, which is a significant performance gain of 46.43% than the baseline method in Table II. Such performance gain underscores the proposed DNN model with meta-learning's enhanced capability in cross-dataset generalization with proposed generalized feature enhancement in the standard ORNL and SAD datasets, as depicted in Table III. Furthermore, in Table III, the proposed DNN model with meta-learning demonstrates an average accuracy of approximately 0.9853 across identical dataset scenarios. This represents an improvement of around 1.79% compared to the baseline model without meta-learning's average accuracy of 0.9679 in Table II. This performance gain in Table III demonstrates the DNN model with meta-learning's enhanced consistency and reliability across various metrics in identical dataset conditions. Similar performance gain is also evident in the case of the proposed RF model with meta-learning in Table IV. In the case of identical dataset scenarios, the average accuracy is approximately 0.9938, and relative to the baseline model without meta-learning in Table II, this indicates a 2.68% increase in accuracy. Besides, the RF model with meta-learning in Table IV shows even greater consistency and performance than the DNN model with meta-learning in Table III in identical scenarios, suggesting significant adaptation to the dataset characteristics. However, regarding the consistency across performance metrics, both the DNN and RF models with meta-learning maintain a higher performance level across various metrics (precision, recall, F1 score) in identical dataset scenarios. This contrasts the baseline model without meta-learning in Table II, which, while showing high accuracy, does not consistently maintain high levels across all metrics. The goal of the proposed dataset generalization is to prevent single-environment over-tuning observed in the baseline on CAN-IDS. Therefore, feature standardization (e.g., *cmafid*, *nuidc*) promotes cross-dataset generalization but down-weights dataset-specific micro-patterns, which can reduce accuracy. This effect is strongest on CAN-IDS: performance dropped in the original cross train–test yet improved under the proposed cross train–test (Tables II–IV). The meta-learning DNN and RF add another layer of generalization that further suppresses micro-patterns, contributing to lower accuracy on the proposed cross train–test (avg). The intended objective is consistency rather than peak accuracy, and the results are validated across scenarios. *Both DNN and RF models with meta-learning outperform the baseline on three of four average settings: identical train–test (avg), proposed-identical train–test (avg), and the original cross train–test (avg), confirming the value of meta-learning and the generalized feature for cross-dataset generalization. In the proposed cross train–test (avg), their scores can be lower than the baseline, yet they consistently exceed the original cross train–test averages (Tables II–IV), reflecting a substantial trade-off that favors robustness across datasets over tuning to a single split. Overall, meta-learning typically improves generalization, with gains dependent on the proposed generalization feature, and the dataset enhancement with preprocessing/augmentation while enhancing model adaptability.*

C. Evaluation of DNN and RF Model Building Process With Proposed Meta-Learning

Fig. 3 illustrates the impact of the proposed meta-learning-based model-building stages for the DNN and RF models within the ACHILLES framework. In Fig. 3a, the RF model demonstrates a narrow interquartile range (IQR), indicating less variability in computation time. Specifically, the IQR occupies about 16.67% of the RF model's computing time scale, suggesting a stable performance, which is crucial for systems like IDS that require consistent processing times. The DNN model, while having a similar proportion of IQR, registers higher absolute values in computation time, with its minimum exceeding the median of the RF model. This suggests that the RF performance under the proposed ACHILLES framework is quite stable and predictable. Such consistency indicates the efficacy of the proposed meta-learning-based model-building process that optimizes the RF model for IDS operations. Besides, the meta-learning with DNN computation time outliers are relatively higher, which indicates the computing time spikes dramatically. It can be explained further as the DNN model requires a substantially higher number of neurons (i.e., mean 200 units in Fig. 3b) and has more hidden layers (i.e., mean 6 hidden layers in Fig. 3c) compared to the RF model's depth (i.e., mean tree depth 25 in Fig. 3d) and the number of trees (i.e., mean number of trees in RF 100 in Fig. 3e). However, despite the higher mean and median, the DNN model's mean does not appear to be excessively influenced by the outliers since it aligns with the median. As a result, the DNN model's typical computing time is reliable with occasional spikes rather than consistently high across all model-building operations. The presence of these outliers in the DNN model indicates that the model faces computational scenarios that are innately more demanding, a common characteristic in DNN tasks that involve large datasets. Overall, the RF model is more suitable for tasks that require fast and consistent processing times, whereas the DNN model is more suitable for complex tasks where higher computing times are acceptable and expected. This comparative analysis assesses the raw computing times and reflects on the nature of tasks each model is likely tackling. It is also noteworthy to mention that the optimized DNN model with meta-learning, despite its slower performance, can deal with operations beyond the optimized RF with meta-learning capabilities, justifying the additional computation time requirements. The RF model's validation accuracy is stable, maintaining above 90% throughout the modeling process, indicating a reliable and consistent model. On the other hand, the DNN model shows higher variability in performance, with accuracy dipping below 85% at times but also reaching peaks that surpass the RF model's accuracy. This suggests that the DNN model may be more sensitive to the specificity of the validation data and could potentially yield higher performance gains in certain scenarios. *Within ACHILLES, the RF model is fast, stable, and reliable as it is well-suited to consistent, time-critical workloads. On the other hand, the DNN is relatively slower with models richer structure and can achieve higher accuracy in some cases. Overall, the meta-learning of ACHILLES exploits these complementary strengths to improve IDS performance.*

D. Feature Analysis With SHAP and Generalizibility

The evaluation of generated feature formats with SHAP for the RF and DNN models are shown in Fig. 4, and in Fig. 6, respectively. We first analyzed the results shown in Figs. 4a and 4c for RF, and Figs. 6a and 6c to understand the global contribution of each SOTA SAD feature to classify non-attack (N-Attack) or attack (Attack). As shown in Figs. 4c, and Figs. 6c, `dt_ID` feature contributed the most to making the final classification decision, where we observe that it is possible to be attacked when the `dt_ID` values are high for both RF and DNN models. Besides, it is likely an attack when the `ID` values are small. Figs. 4b and 4d, and Figs. 6b and 6d provide the realization of the global contribution of each proposed SAD feature to classify non-attack (N-Attack) or

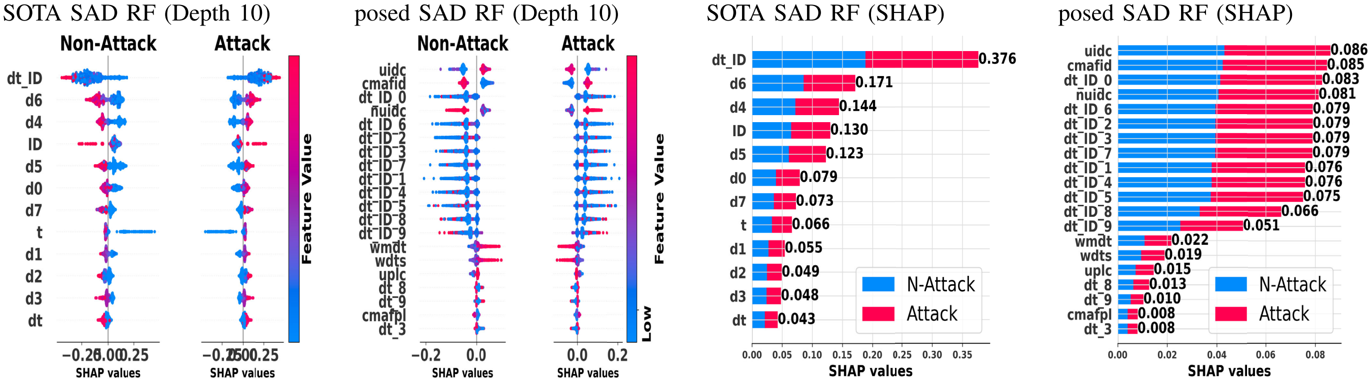


(a) Global SHAP values for SOTA SAD RF (Depth 10)

(b) Global SHAP values for Proposed SAD RF (Depth 10)

(c) Features importance for SOTA SAD RF (SHAP)

(d) Features importance for Proposed SAD RF (SHAP)



(e) Global SHAP values for SOTA ROAD RF (Depth 10)

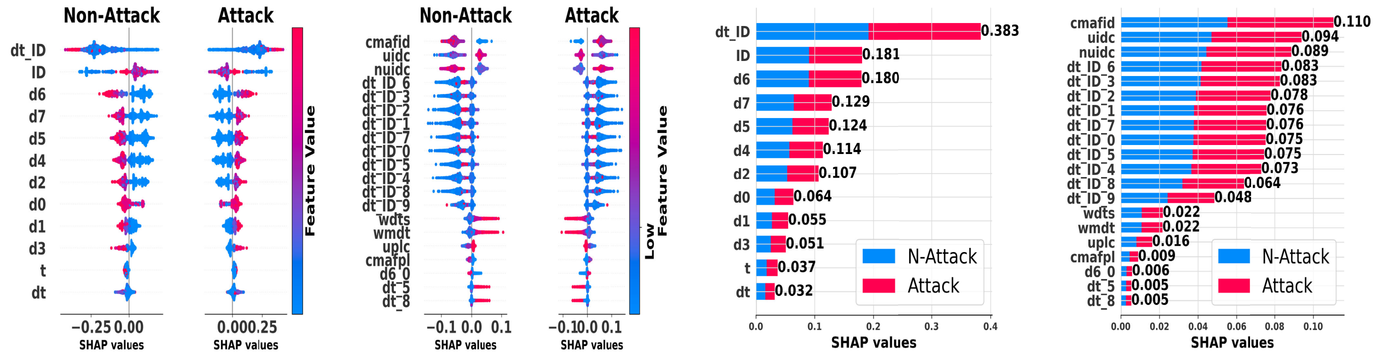
(f) Global SHAP values for Proposed ROAD RF (Depth 10)

(g) Features importance for SOTA ROAD RF (SHAP)

(h) Features importance for Proposed ROAD RF (SHAP)

Fig. 4. Evaluation of generated feature formats with SHapley additive exPlanations (SHAP) for random forest model.

attack (Attack). As shown in Fig. 4d and Fig. 6d, the `cmafid` feature contributed the most to making the final classification decision, where we observe, it is possible to be attacked when the `cmafid` along with the `nuidc`, `cmafpl` values are high. As for the `upic` and `uidc`, it is possible to be attacked when the `upic` and `uidc` values are small. Therefore, the users can easily understand the decision of the Proposed SAD RF and DNN models based on Figs. 4b and 4d, and Figs. 6b and 6d, respectively. Then, we analyzed the results shown in Figs. 4e and 4g, and Figs. 6e and 6g to illustrate the global contribution of each SOTA ROAD feature to classify non-attack (N-Attack) or attack (Attack). As shown in Fig. 4g, and Fig. 6g, we can see that the `d3` contributed the most to make the final classification decision, where we can see that it is possible to be attacked when the `d3` along with the `d1`, `d0` values are high. Therefore, the decision of the SOTA ROAD RF, as well as DNN models, can be easily explained based on Figs. 4e and 4g, and Figs. 6e and 6g, respectively. Finally, we analyzed the results shown in Figs. 4f and 4h, and Figs. 4f and 4h to understand the global contribution of each Proposed ROAD feature to classify non-attack non-attack (N-Attack) or attack (Attack). As shown in Figs. 4h, we can see that the `ID_2` contributed the most to make the final classification decision, where we can see similar to Fig. 6h. Based on the Figs. 4f and 6f, it is possible to be attacked when the `cmafid` along with the `nuidc`, `cmafpl` values are high. As for the `ID_2` and `ID_3`, `ID_0` and `dt_ID_1`, it is possible to be attacked when those values are small. Again, the relevant users can easily understand the decision of the Proposed ROAD RF as well as DNN models based on Figs. 4f and 4h, and Figs. 6f and 6h. Fig 5 and 7 provide a comparison of SHAP value analyses for Random Forest (RF) and Deep Neural Network (DNN) models on CAN-IDS and CAR-HACK datasets. They offer a clear view of how each feature influences the model's predictions for both non-attack and attack classifications. Fig 5a shows the global SHAP values for the SOTA CAN-IDS RF model. The distribution for non-attack and attack classifications suggests that certain features like `d0` and `d1` have SHAP values that significantly influence the model's predictions, extending past 0.25 for attacks. Fig 5c emphasizes feature importance, with the highest SHAP value for non-attacks at approximately 0.383 for ID, indicating its significant impact on the RF model. Fig 7a for the CAN-IDS DNN model reveals that features such as `d6` and `d5` display SHAP values with wide distributions for both non-attack and attack classes, indicating a variable impact on the model's predictions. Fig 7c highlights feature importance for CAN-IDS DNN, with the highest SHAP value for attacks at roughly 0.368 for `nuic`, which shows its critical role in the DNN model. Fig 5e presents a tightly clustered distribution of SHAP values for the SOTA CAR-HACK RF, suggesting a more consistent feature impact across predictions. Fig 5g shows a strong feature importance for non-attacks, with `d7` having a SHAP value of about 0.199, making it highly influential in the RF model. Fig 7e for the CAR-HACK DNN model indicates a wider spread of SHAP values for features like `d7` and `d3`, highlighting their variable

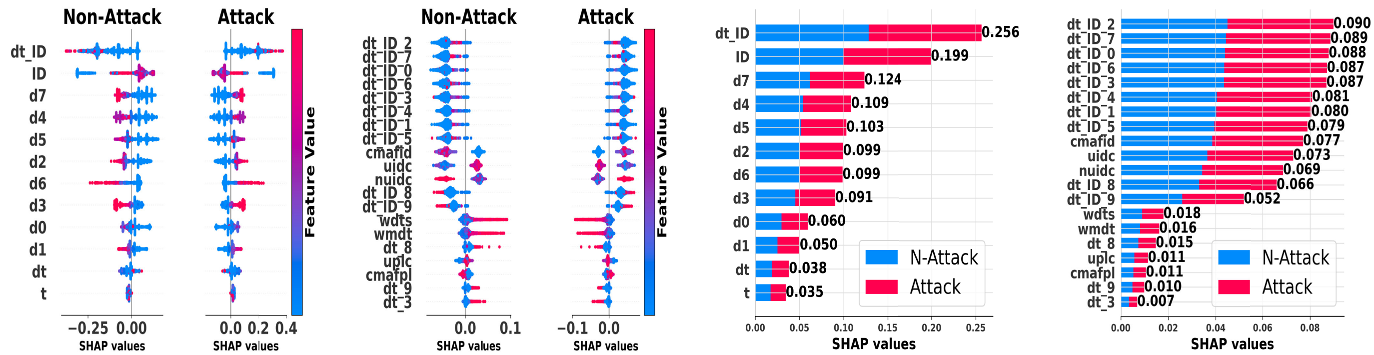


(a) Global SHAP values for SOTA CAN-IDS RF (Depth 10)

(b) Global SHAP values for Proposed CAN-IDS RF (Depth 10)

(c) Features importance for SOTA CAN-IDS RF (SHAP)

(d) Features importance for Proposed CAN-IDS RF (SHAP)



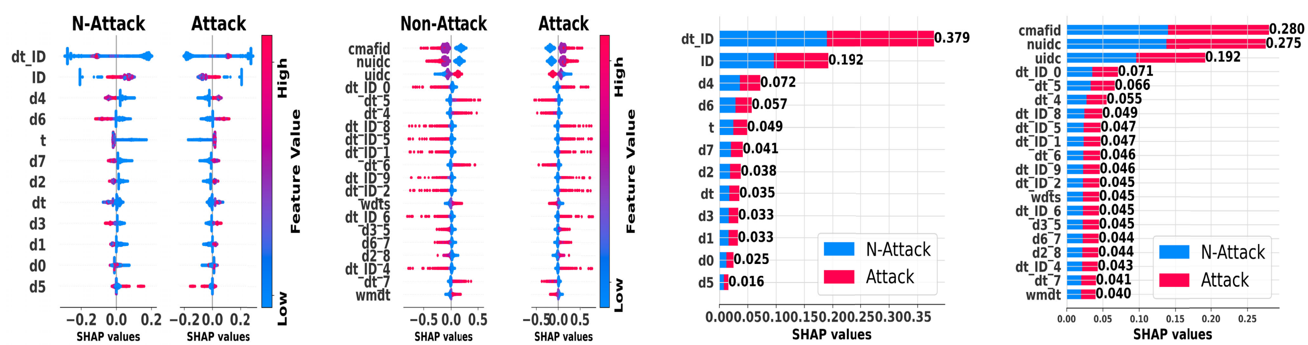
(e) Global SHAP values for SOTA CAR-HACK RF

(f) Global SHAP values for Proposed CAR-HACK RF

(g) Features importance for SOTA CAR-HACK RF (SHAP)

(h) Features importance for Proposed CAR-HACK RF (SHAP)

Fig. 5. Evaluation of generated feature formats with SHapley additive exPlanations (SHAP) for random forest model.

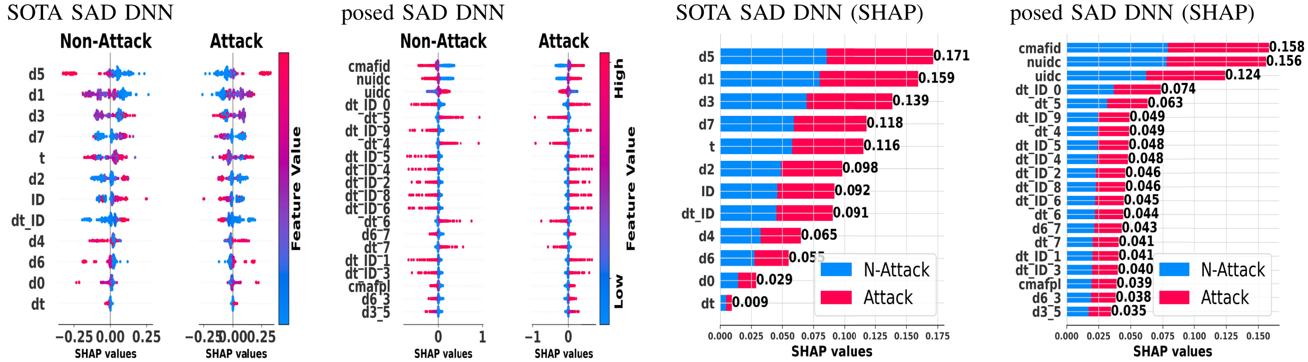


(a) Global SHAP values for SOTA SAD DNN

(b) Global SHAP values for Proposed SAD DNN

(c) Features importance for SOTA SAD DNN (SHAP)

(d) Features importance for Proposed SAD DNN (SHAP)



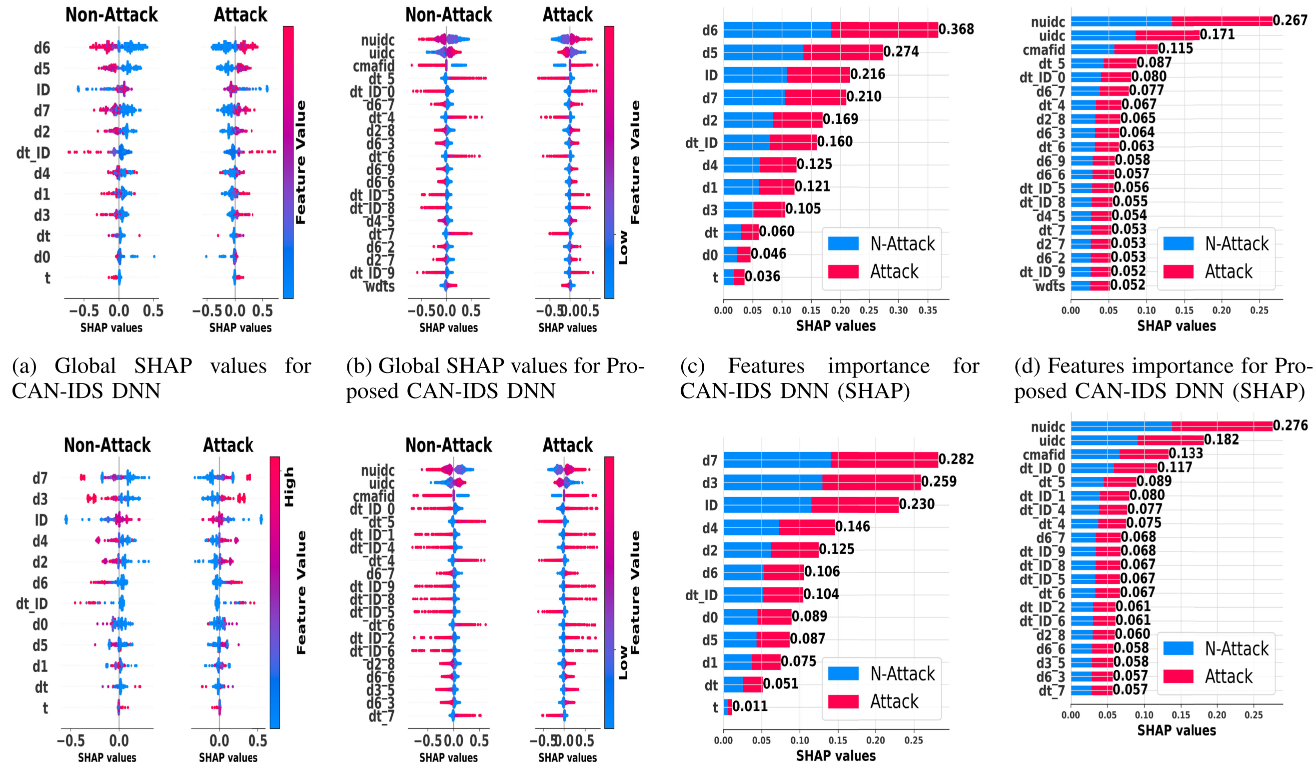
(e) Global SHAP values for SOTA ROAD DNN

(f) Global SHAP values for Proposed ROAD DNN

(g) Features importance for SOTA ROAD DNN (SHAP)

(h) Features importance for Proposed ROAD DNN (SHAP)

Fig. 6. Evaluation of generated feature formats with SHapley additive exPlanations (SHAP) for DNN model.



(a) Global SHAP values for CAN-IDS DNN

(b) Global SHAP values for Proposed CAN-IDS DNN

(c) Features importance for CAN-IDS DNN (SHAP)

(d) Features importance for Proposed CAN-IDS DNN (SHAP)

(e) Global SHAP values for CAR-HACK DNN

(f) Global SHAP values for Proposed CAR-HACK DNN

(g) Features importance for CAR-HACK DNN (SHAP)

(h) Features importance for Proposed CAR-HACK DNN (SHAP)

Fig. 7. Evaluation of generated feature formats with SHapley additive exPlanations (SHAP) for DNN model.

influence on the model's predictions. Fig 7g displays the highest SHAP value for non-attacks at approximately 0.282 for `nuic`, underlining its paramount importance in the DNN model.

1) Generalizability Analysis: After recognizing the feature characteristics utilized to make final classification decisions, we examine why the generalizability of the proposed approaches outperforms SOTA SAD and SOTA ROAD in terms of generalization performance. First, we analyze Figs. 4a and 4e to understand the decisions made by SOTA SAD RF and SOTA ROAD RF. As shown in the figure, common features such as `d0`, `d5`, `d7` have opposite contributions to the model's decision. Second, we analyze Figs. 6a and 6e to understand the decisions made by SOTA SAD DNN and SOTA ROAD DNN. As shown in the figure, common features such as `dt_ID`, `d3`, `5` have opposite contributions to the model's decision. Third, we analyze Figs. 4b and 4f to understand the decisions made by Proposed SAD RF and Proposed ROAD RF. As shown in the figure, tailored features such as `cmafid`, `nuidc`, `uidc` and `dt_ID_0` have similar contributions to the model's decision. Fourth, we analyze Figs. 6b and 6f to understand the decisions made by Proposed SAD RF and Proposed ROAD RF. Tailored features such as `nuidc` and `uidc` contribute similarly to the model's decision. Therefore, if the global SHAP values have similar contributions, we observe that the models indicate significant performance gain for cross-data. Meanwhile, in the case of Fig. 5 and Fig. 7, the high importance of a few features, such as `ID`, indicates that the RF model's decision-making is significantly influenced by these features, which may affect generalizability when the importance of these features is specific to the CAN-IDS training data as shown in Figs. 5a–5d. On the other hand, the DNN model results in Figs. 7a–7d show a wider spread of SHAP values, indicating a more complex interaction between features and the model's predictions. This signifies better generalizability as the model does not rely on a small set of features. The consistent importance of features like `d7` across non-attack and attack classes suggests the RF model may generalize well when these features hold similar significance in unseen CAR-HACK data, as shown in Figs. 5e–5h. In the case of the DNN model with the CAR-HACK dataset in Figs. 7e–7h, the model's SHAP values are more spread out, potentially reflecting a model that has learned a richer set of patterns that could generalize well to new data. The `dt_ID` feature holds particular importance as it appears prominently across the SHAP value plots for both the RF and DNN models on CAN-IDS and CAR-HACK datasets. In the SOTA CAN-IDS RF model in Fig. 5c, `dt_ID` demonstrates substantial importance for non-attack classifications with a SHAP value of 0.383, making it one of the top predictors. In the CAN-IDS DNN model (Fig. 7c), `dt_ID` also shows up as a key feature for attacks, with a SHAP value of 0.160, indicating its significant role in the DNN's classification process. For the proposed CAN-IDS DNN model (Fig. 7d), `dt_ID` continues to be an important feature but is less dominant compared to the SOTA model.

V. Discussion

ACHILLES targets deployment via *centralized training* and *decentralized on-vehicle inference* where the proposed meta-learning runs off-board to select compact models and only final weights are pushed to minimize ECU load. As shown in Section IV-B, RF achieves ~0.03–0.04,s per evaluation window, enabling real-time use on gateway/security ECUs (e.g., C400-class). The inference is windowed (n=10 frames) and non-blocking on the gateway/security ECU. The measured ~40 ms per-window inference *does not introduce blocking latency* for actuators. Instead, it bounds the time to *detect and respond* at the CAN bus level (e.g., filtering/rate-limiting an offending ID or isolating a sender at the gateway) without delaying nominal ECU processing. Prior work supports lightweight embedded IDS, which is, RF on real CAN attacks [55] and compact DNNs with sub-ms latency [56]. Cross-vehicle testing is the stress test: standardized timing/frequency features enable a vendor-agnostic IDS to scale across mixed fleets, share abstract statistics instead of proprietary payloads, push central updates rapidly, and keep SHAP explanations consistent. All heavy computation (meta-search/retraining) remains off-board, preserving portability and transparency.

VI. Conclusion

This paper proposes an explainable and generalizable framework for an automotive IDS that leverages centralized machine learning model training for decentralized model deployment. We have generated augmented features and developed a meta-learning algorithm for automotive IDS model building. The proposed ACHILLES improves the generalizability of the models that are tested with multiple feature sets using deep neural networks and random forest models leveraging meta-learning-based model building and evolution process. Also, the proposed framework is evaluated using the XAI technique to identify the most contributing feature sets. Our XAI evaluation validate the explanation of the generalization and create trust in the automotive IDS by demonstrating that the proposed features lead to improved performance in a variety of tests and evaluation metrics. The performance evaluation shows that, under ACHILLES, the proposed generalized feature format together with tuned model parameters jointly improves cross-dataset automotive attack classification. While ACHILLES demonstrates clear gains in cross-vehicle generalisation and explainability, our feature set is CAN-centric. In future, we will extend the feature-normalisation pipeline to mixed-protocol logs to extend the protocol scope for IVN. We also plan to investigate adversarial poisoning and federated defenses.

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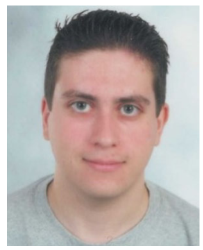

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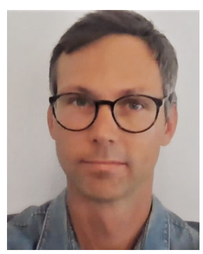

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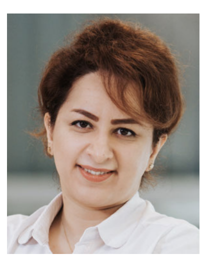

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